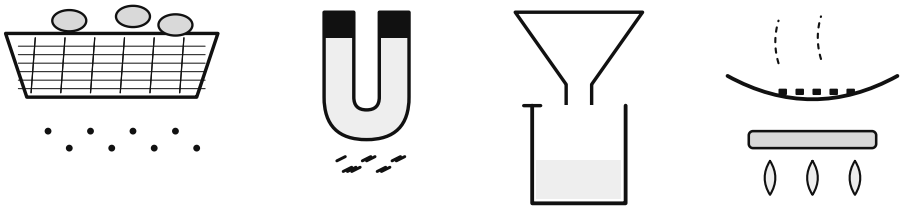
PHYSICAL AND CHEMICAL SCIENCE

# Mixtures, Solutions and Changes

Separating mixtures, dissolving and solutions, reversible and irreversible changes



**sieving** **using a magnet** **filtering** **evaporating**

**In this pack**

- **Learn it:** Mixing, dissolving, separating and changing ★☆☆
- **Fact check:** Separating and changing ★☆☆
- **Think it through:** Solving separation problems ★★☆
- **Investigate:** Does warm water dissolve sugar faster? ★★☆
- **Real-world case study:** Khejur gur: winter sweetness and a hidden danger ★★★
- **Review and challenge:** Show what you know ★★★

Also covers Cambridge Primary Science Stage 5: Unit 3 States of matter: 3.4 Investigating evaporation from a solution.

**This pack belongs to:** _____ **Class:** _____

© InquiryMind · Original learning material for Grade 5 (PHIS Science continuum, IB PYP Phase 4, and Cambridge Primary Science Stage 5) · Illustrations drawn for InquiryMind · Print in black and white on A4 · Challenge: ★ core ★★ deeper ★★★ stretch

# Mixing, dissolving, separating and changing

★☆☆ Grade 5

Name: Class: Date:

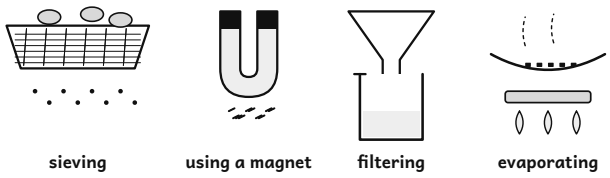
**Learning goals**

- Separate mixtures by sieving, filtering, using a magnet and evaporating.
- Explain dissolving and why a dissolved substance is still there.
- Tell the difference between reversible and irreversible changes.

Think about a cup of cha: tea leaves, water, milk and sugar. Some things mix and can be separated again; some dissolve and seem to disappear; and some changes, like cooking an egg, can never be undone.

## Mixtures

A **mixture** is two or more materials mixed together but not joined. Rice with small stones, sand with water, and iron nails with sawdust are all mixtures. Because the materials keep their own properties, we can use those properties to **separate** them.




*Four ways to separate mixtures.*

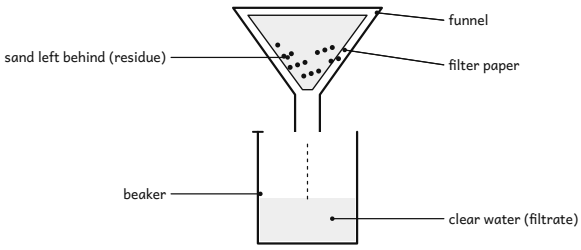
- **Sieving:** separates solid pieces of different sizes, for example stones from rice, or lumps from flour.
- **Using a magnet:** pulls out magnetic materials such as iron and steel.
- **Filtering:** separates an **insoluble** solid (one that does not dissolve) from a liquid, for example sand from water. The liquid passes through tiny holes in the filter paper; the solid stays behind. It is like a very fine sieve.
- **Evaporating:** gets a **dissolved** solid back from a solution. The water evaporates; the solid is left.

**Bangladesh connection**

Farmers use a flat bamboo tray called a **kula** to separate rice from its light husks. They toss the rice in the breeze: the heavy grains fall back, and the wind carries the light husks away.

## Dissolving

When you stir sugar into water, the sugar seems to disappear. It has **dissolved**: it has broken into pieces far too small to see and spread through the water to make a **solution**. The sugar is still there. You can taste it, and the mass of the solution equals the mass of the water plus the mass of the sugar. Materials that dissolve are **soluble** (salt, sugar); materials that do not are **insoluble** (sand, chalk, flour). Stirring, using **warmer** water and using **smaller** pieces make things dissolve faster.




*Filtering sandy water.*

### Reversible and irreversible changes

A **reversible change** can be undone: melting and freezing, dissolving (evaporate the water to get the solid back), and mixing and separating. No new material is made.

An **irreversible change** cannot be undone, because **new materials** are made: cooking an egg, baking a cake, burning wood or paper, iron **rusting**, and cement setting. Signs of a new material include bubbles of gas, a colour change, a new smell, and heat or light being given out.

> ⚠ **Stay safe**
> Never taste anything in a science lesson, and ask an adult to help with heating.

📖 **Key words**

| | |
|---|---|
| **mixture** | materials mixed together but not joined |
| **dissolve** | break up and spread through a liquid |
| **solution** | a liquid with something dissolved in it |
| **soluble** | able to dissolve |
| **insoluble** | not able to dissolve |
| **filtering** | separating an insoluble solid from a liquid |
| **reversible change** | a change that can be undone |
| **irreversible change** | a change that makes new materials and cannot be undone |

# Separating and changing



Name: Class: Date:

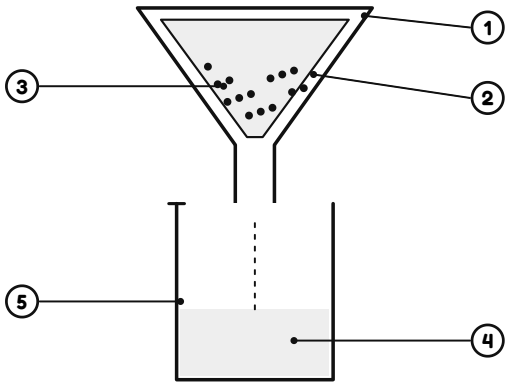
**Learning goals**

- Choose a separation method.
- Sort reversible and irreversible changes.

**1** **Label the filtering equipment.** [5]

LABELS

**funnel** **filter paper** **clear water (filtrate)** **sand left behind (residue)** **beaker**




1 _____ 2 _____

3 _____ 4 _____

5 _____

**2** **Which method would you use to separate each mixture?** [4]

| Mixture | Method |
|---|---|
| small stones from rice | |
| iron filings from sand | |
| sand from water | |
| salt from salty water | |

**3** **Sort the materials.** [6]

SORT THESE

**salt** **sugar** **sand** **flour** **instant coffee** **chalk**

| Soluble in water | Insoluble in water |
|---|---|
| | |
| | |
| | |

**4** **Sort the changes.** [6]

SORT THESE

**sugar dissolving** **ice melting** **frying an egg** **a nail rusting** **water freezing** **burning paper**

| Reversible | Irreversible |
|---|---|
| | |
| | |
| | |

# Solving separation problems

★★☆
Grade 5

Name: Class: Date:

**Learning goals**

- Plan how to separate a complex mixture.
- Explain dissolving and new materials.

**1** **Rana knocked over a jar. Now sand, salt and iron nails are all mixed together. Explain step by step how to get all three back.** [4]

**2** **Sumi says, "When sugar dissolves it disappears, so it is gone." Use evidence to explain why she is wrong.** [2]

**3** **How can you tell that baking a cake makes a new material? Give two signs.** [2]

**4** **Why does filtering not separate salt from salty water?** [2]

# Does warm water dissolve sugar faster?

★★☆ Grade 5

Name: Class: Date:

**Learning goals**

- Identify variables.
- Plot and read a line graph.
- Make a prediction from a pattern.

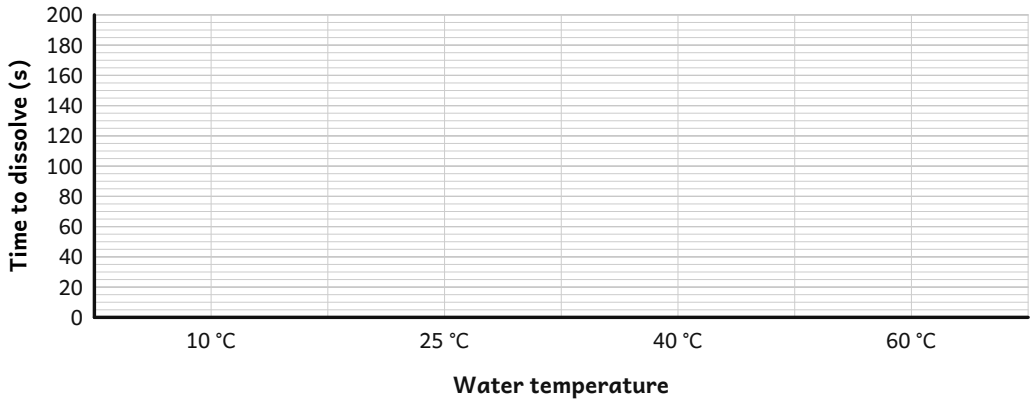
Class 5B timed how long one teaspoon of sugar took to dissolve in 100 ml of water at different temperatures, stirring the same way each time.

**1** **Name the variables.** [3]

| | |
|---|---|
| Change | |
| Measure | |
| Keep the same | |

**2** **Plot the results as a line graph.** [4]

| Water temperature (°C) | 10 | 25 | 40 | 60 |
|---|---|---|---|---|
| Time to dissolve (s) | 180 | 95 | 50 | 25 |




**3** **Describe the pattern.** [2]

.....

.....

**4** **Predict the time at 50 °C. Explain how you decided.** [1]

.....

.....

**5** **How could they make the results more reliable?** [1]

.....

.....

REAL-WORLD CASE STUDY · C2 · MIXTURES AND CHANGES

# Khejur gur: winter sweetness and a hidden danger

★★★ Grade 5

Name: Class: Date:

**Learning goals**

- Identify reversible and irreversible changes in a real process.
- Explain a health risk using science.
- Communicate a safety message.

Science · Health · Culture · Economics

On cold winter nights in Jashore and other districts, skilled climbers called *gachhi* cut the bark of date palm trees and hang clay pots to collect the sweet **sap**. In the morning the sap is **filtered** through cloth to remove insects and bits of bark, then **boiled** for hours in wide pans. Water **evaporates**, the sap gets thicker, and the sugar turns golden brown. Poured into moulds, it sets as **khejur gur** (date palm molasses), used to make pitha and payesh.

But raw, unboiled sap can be dangerous. Fruit bats sometimes drink from the pots at night and can leave behind the **Nipah virus**, which causes a serious brain illness. Health experts in Bangladesh warn: **never drink raw date palm sap**. Boiling kills the virus, so properly made gur is safe.

Health advice from IEDCR (Institute of Epidemiology, Disease Control and Research), Bangladesh.

**1** **Find the separating methods and changes in the story. Complete the table.** [8]

| Step | What happens | Reversible or irreversible? |
|---|---|---|
| filtering through cloth | | |
| boiling: water evaporates | | |
| sugar turns golden brown | | |
| gur sets in moulds | | |

**2** **Why is filtering through cloth not enough to make raw sap safe to drink?** [2]

.....

.....

**3** **Write a short safety message (radio or poster) for families in your area about date palm sap.** [2]

My message

# Show what you know

★★★
Grade 5

Name: Class: Date:

**Learning goals**

- Use key vocabulary.
- Answer test-style questions.

**1** **Find the key words.**

| J | B | F | W | M | I | U | R | D | T | T |
|---|---|---|---|---|---|---|---|---|---|---|
| P | U | B | O | A | L | S | N | M | S | S |
| I | R | K | J | G | S | O | C | I | I | O |
| W | N | A | F | N | I | L | R | X | E | L |
| V | A | G | I | E | V | U | P | T | V | U |
| S | J | U | L | T | H | T | I | U | E | B |
| O | A | E | T | R | U | I | A | R | R | L |
| M | O | G | E | G | Y | O | I | E | U | E |
| H | N | R | R | R | V | N | Y | K | S | M |
| B | W | S | N | O | E | R | M | O | T | O |
| O | N | D | I | S | S | O | L | V | E | W |

**burn** **dissolve** **filter** **magnet** **mixture** **rust** **sieve** **soluble** **solution**

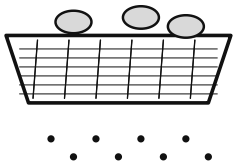
**2** **Which of these is an irreversible change?** [1]

- ◯ melting chocolate
- ◯ dissolving salt
- ◯ burning a match
- ◯ freezing juice

**3** **Which method separates sand from water?** [1]

- ◯ evaporating
- ◯ filtering
- ◯ using a magnet
- ◯ dissolving

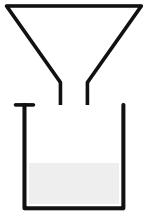
**4** **Write the letter of each separation method.** [4]



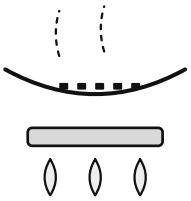
A



B



C



D

| Method | Letter |
|---|---|
| filtering | |
| evaporating | |
| sieving | |
| magnet | |

**5** **Challenge. Is making a cup of cha with sugar and milk a reversible or irreversible change? Explain your thinking.** [2]

.....

.....

.....

**How am I doing? Colour or tick a circle.**

| I can... | Not yet | Nearly | Got it! |
|---|---|---|---|
| choose a method to separate a mixture | ○ | ○ | ○ |
| explain filtering and why it is like sieving | ○ | ○ | ○ |
| explain that a dissolved solid is still there | ○ | ○ | ○ |
| sort changes into reversible and irreversible | ○ | ○ | ○ |
| investigate what makes things dissolve faster | ○ | ○ | ○ |
| explain why raw date palm sap is unsafe | ○ | ○ | ○ |